

Technological Approaches for Adaptation and Personalization in Educational Video Games

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Abstract. Nowadays, educational video games are a modern means of implementing various pedagogical practices and learners of different ages and educational levels are trained daily through them. There are various genres of educational video games that combine educational content with engaging gameplay, and this paper presents a brief overview of the most popular of them. The most often applied approaches for game personalization and adaptation are also described. The authors conducted both quantitative and qualitative research on the studied area. A systematic quantitative review was performed according to the PRISMA (Preferred Reporting Items for Approaches for Reviews and Meta-Analyses) methodology, providing a comprehensive overview of publications related to the adaptation and personalization approaches applicable to educational video games. The qualitative research is conducted using a software tool, which enables content analysis of the results obtained through the quantitative research methodology. The final dataset included 246 scientific publications related to approaches for tailoring educational video games, published in the last 10 years in the Web of Science and Scopus databases. The authors derived information on the most popular approaches for describing and managing adaptation and personalization in educational video games across various learning domains and current trends of their evolution.